

THESSALONIKI MAKEDONIA, Greece

WMO: 166220

Lat: 40.520N

Lon: 22.971E

Elev: 7

StdP: 101.24

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -3.0	(c) -1.2	(d) -9.9	(e) 1.6	(f) 0.3	(g) -7.9	(h) 1.9	(i) 1.5	(j) 14.7	(k) 6.6	(l) 12.8	(m) 5.2	(n) 3.5	(o) 110	(p) 0.439

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.2	34.9	21.7	33.2	21.5	32.1	21.2	23.8	31.1	23.0	30.4	22.4	29.6	4.7	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
21.2	15.9	28.3	20.4	15.0	27.5	19.9	14.6	27.0	71.1	31.2	68.2	30.5	65.7	29.6	27.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 11.1	(b) 9.2	(c) 7.9	(e) -5.5	(f) 38.2	(g) 2.0	(h) 2.0	(i) -7.0	(j) 39.7	(k) -8.1	(l) 40.9	(m) -9.3	(n) 42.0	(o) -10.7	(p) 43.5		
(a) 11.1	(b) 9.2	(c) 7.9	(e) -6.7	(f) 25.5	(g) 1.9	(h) 1.1	(i) -8.0	(j) 26.3	(k) -9.2	(l) 26.9	(m) -10.2	(n) 27.5	(o) -11.6	(p) 28.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	16.1	5.8	7.3	10.1	14.2	19.5	24.3	26.7	26.6	22.0	16.8	11.8	7.2		
	DBStd	7.99	3.43	3.40	3.16	2.83	2.73	2.64	2.04	1.97	2.59	3.13	3.59	3.69		
	HDD10.0	387	136	86	37	3	0	0	0	0	0	1	22	102		
	HDD18.3	1711	389	309	255	127	19	0	0	0	3	65	197	346		
	CDD10.0	2606	4	10	41	129	296	430	518	516	360	211	76	15		
	CDD18.3	887	0	0	0	2	56	180	259	257	113	17	1	0		
	CDH23.3	8320	0	0	3	30	404	1636	2786	2642	747	70	2	0		
	CDH26.7	3321	0	0	1	3	83	634	1236	1151	205	7	0	0		
(14) Wind	WSAvg	2.9	2.9	2.9	3.0	2.8	2.8	3.2	3.3	3.0	2.9	2.6	2.6	2.9		
(15) Precipitation	PrecAvg	448	37	36	41	34	43	29	24	20	33	41	50	55		
	PrecMax	734	111	167	106	108	98	86	93	98	141	194	179	164		
	PrecMin	236	0	0	2	3	0	0	0	0	0	0	6	0		
	PrecStd	96	26	30	26	26	25	23	23	18	34	37	35	40		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.0	18.2	22.1	26.5	31.0	35.9	37.2	36.8	32.9	27.8	22.9	19.2		
		MCWB	11.8	11.6	13.1	15.9	18.6	21.5	21.7	21.6	20.1	19.5	16.2	14.0		
	2%	DB	14.2	16.1	19.2	23.9	28.9	33.3	35.0	34.8	30.8	25.1	20.2	16.8		
		MCWB	10.4	10.9	12.1	15.2	18.1	21.3	21.6	21.9	19.9	18.0	15.7	12.7		
	5%	DB	12.2	14.2	17.5	21.9	27.2	31.9	33.4	33.1	29.0	23.5	18.9	14.2		
		MCWB	9.3	10.2	11.5	14.1	17.6	20.7	21.5	21.7	19.6	17.3	15.0	11.3		
	10%	DB	10.9	12.9	15.8	20.0	25.8	30.2	32.0	32.0	27.2	22.0	17.2	12.8		
		MCWB	8.5	9.4	11.0	13.6	17.2	20.4	21.3	21.4	19.4	16.8	14.0	10.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.1	13.0	14.9	17.6	20.9	24.1	24.7	25.1	22.5	20.8	17.7	15.0		
		MCDB	15.8	16.3	18.8	23.1	27.9	31.3	32.5	32.2	28.5	25.3	20.5	19.0		
	2%	WB	11.0	11.7	13.3	16.2	19.7	22.8	23.7	24.0	21.5	19.4	16.5	13.2		
		MCDB	13.2	14.8	17.2	21.4	26.0	30.3	31.2	31.1	27.3	23.3	19.3	15.8		
	5%	WB	10.0	10.7	12.3	15.2	18.8	22.0	22.9	23.1	20.8	18.4	15.5	11.9		
		MCDB	11.7	13.6	16.2	19.9	24.7	29.2	30.3	30.5	26.5	22.0	18.3	13.9		
	10%	WB	8.9	9.9	11.4	14.3	18.0	21.1	22.3	22.4	20.2	17.5	14.5	10.7		
		MCDB	10.4	12.2	15.0	18.6	23.7	28.4	29.7	29.6	25.8	20.8	17.0	12.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.2	8.0	9.3	10.2	11.0	11.1	11.2	11.0	10.1	8.7	7.8	7.1		
		MCDBR	9.4	10.7	12.7	13.5	13.3	13.1	13.0	12.7	12.4	11.2	9.1	8.8		
	5% WB	MCWBR	6.7	7.0	7.3	7.0	6.1	5.9	6.1	6.1	6.2	6.2	5.9	6.0		
		MCWBR	8.0	9.4	11.2	11.5	11.9	12.1	11.9	11.7	11.3	9.4	7.9	7.1		
(40) Clear-Sky Solar Irradiance	taub	taub	0.359	0.390	0.433	0.477	0.465	0.459	0.459	0.471	0.451	0.425	0.392	0.368		
		taud	2.363	2.242	2.130	2.030	2.139	2.199	2.204	2.164	2.193	2.237	2.279	2.325		
	Ebn at Noon	Ebn at Noon	784	808	810	800	820	824	820	797	784	758	731	735		
		Edh at Noon	87	113	140	166	152	143	141	143	131	112	94	84		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.80	2.60	3.79	5.01	6.12	6.93	7.15	6.34	4.68	3.13	2.06	1.61		
	RadStd	RadStd	0.26	0.33	0.38	0.52	0.30	0.40	0.29	0.29	0.34	0.37	0.24	0.27		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.42	N/A	N/A	N/A	N/A	N/A	N/A	-102	N/A	N/A
(47) Regional (17 neighbors)		+0.38	N/A	N/A	N/A	N/A	N/A	N/A	-114	N/A	N/A

Nomenclature: See separate page